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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

ECH618 – POLYMER CHEMISTRY

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. Which of the following is not a naturally occurring polymer?
   a) Cellulose b) Starch c) DNA d) Terylene

2. The third most widely used polymer is
   a) Polystyrene b) Polyethylene
   c) Polypropylene d) PVC

3. Caprolactum is the monomer for
   a) Nylon-66 b) Nylon-6 c) Epoxide resin d) Glyptal

4. Which of the following has an ester linkage?
   a) Bakelite b) Nylon c) Terylene d) PVC

5. The transition of glassy solid into molten liquid is called
   a) Critical temperature b) Boyle temperature
   c) Glass transition temperature d) Inversion temp erature

6. Filament-winding technique is used in
   a) Foaming b) Moulding c) Spinning d) Reinforcing

7. For all synthetic polymers,the number & weight average molecular weights are related as
   a) $M_w<M_n$ b) $M_w>M_n$ c) $M_w=M_n$ d) $M_w=M_n^2$

8. The ratio of number and weight average molecular mass of polymer is called
   a) Degree of crystallinity b) Degree of polymerization
   c) Poydispersity index d) Depropagation

9. Recycle number 5 stands for the polymer
   a) PVC b) Polypropylene c) PET d) Polystyrene

10. Poly amide is
    a) Bakelite b) Epoxy resin c) Silicones d) Nylon

II. Fill up the blanks

11. Mathematically, weight average molecular mass of polymer is given as _____

12. The degree of polymerization and molecular mass of a polymer are related as _____

13. HDPE is _____

14. The heterogeneity of a polymer sample is called_____

15. A polymer of phenol with excess formaldehyde is called_____

III. Match the following

16. Terylene	a) Abrasion resistance
17. RNA	b) Adipic acid
18. Polymer testing	c) Glycol
19. Nylon 66	d) Grafting
20. Branched polymer	e) Biopolymer

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. Calculate the polydispersity index, if mean average mass and number average masses are 78,005 and 43,100.

22. How nylon-6 and polystyrene are prepared?

23. How polymers are classified on the basis of molecular forces.

24. Draw the plot of glass transition temperature and molecular weight of a polymer.

25. What are plastics? Give an example.

26. Equal masses of polymer (say 200,000) molecules with M_1 = 100,000 and M_2 = 10,000 are mixed. Calculate number average molar mass.

27. How T_g is related with T_m for symmetrical and unsymmetrical polymers?

28. Write about the strain induced morphology of crystalline polymer.

29. What are network polymers? Give one example.

30. How phenolic resins are prepared?

31. Write about fire retarding polymers.

32. How a cross-linked silicone polymer is prepared from alkylchlorosilane.

SECTION – C (5X7=35)

Answer any **TEN** Questions

33. a) Write about polymerization in homogeneous system. **(3)**

b) 20 numbers of molecules with M_1 = 50,000 and M_2 = 20,000 are mixed, calculate the number and mass average molar masses. **(2+2)**

34. a) Briefly write about morphology and order in crystalline polymers. **(2+2)**

b) Write about strain induced morphology of crystalline polymers. **(3)**

35. a) Write schematic diagram explaining thermo-forming process. **(3)**

b) What is reinforcing? Draw schematic diagram showing spray-up technique. **(2+2)**

36. a) Write about hardness and abrasion resistance of polymers. **(2+2)**

b) The intrinsic viscosity of a solution of polyisobutylene at 20^0 C is 180$cm^3.g^{-1}$. If $[\eta]$ is related to the viscosity-average molar mass $< M_{vis} >$ by the expression. $[\eta] = 3.60X10^{-2}$ $(< M_{vis} >)$, calculate the molar mass of the polymer. **(3)**

37. a) Give an example for biomedical polymer used in artificial heart and contact lens. **(2)**

b) Briefly discuss about electrically conducting polymers. **(3)**

c) What is Number average molecular weight of a polymer? **(2)**

38. a) Write the mechanism of condensation polymerization. **(3)**

b) Write about linear and branched polymers. **(3)**

39. a) Define fibers.Give two examples. **(3)**

b) Name the monomers for **(1+1+1+1)**

i) SAN copolymer ii) ABS terpolymer

iii) PAN polymer iv) PMMA

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